BATCH & BUILD · STEM LEARNING SHEET · AGES 8–12

Crystal Growing: Building Shapes Atom by Atom

THE SCIENCE

Hot water can hold more dissolved stuff, like salt, sugar, or borax, than cold water. When you stir in so much that no more will dissolve, the solution is saturated. As it cools and the water slowly evaporates, there isn't room for all of it, so the tiny particles start joining back together. They line up in a neat, repeating pattern, and that pattern is a crystal. Each material has its own crystal shape: table salt makes cubes.

The slower crystals grow, the bigger and neater they get. That's why some of the biggest crystals on Earth formed over thousands of years in quiet caves.

KEY WORDS

Dissolve: when a solid breaks into tiny pieces and mixes into a liquid.

Solution: a liquid with something dissolved in it.

Saturated: holding as much dissolved material as it can.

Evaporate: when liquid turns into a gas and floats away.

Crystal: a solid with particles in a repeating pattern.

Seed crystal: a tiny crystal that bigger crystals grow on.

EXPERIMENT – FAST OR SLOW?

Make one saturated salt solution and split it into two jars. Put one jar in the fridge and leave the other on a quiet shelf. Hang a string in each. After 3 days, compare crystal size with a magnifying glass.

Think about it: Which grew bigger crystals, and why?

SAFETY

An adult handles the hot water. If the kit uses borax, don't taste it, wash hands after, and keep it away from young kids and pets.

CHECK YOUR UNDERSTANDING

1. What does saturated mean?
2. Does hot or cold water dissolve more?
3. What shape do salt crystals make?
4. Do slow-growing crystals end up bigger or smaller?
5. What is a seed crystal?

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ANSWER KEY (FOR GROWN-UPS)

1. It can't dissolve any more.
2. Hot.
3. Cubes.
4. Bigger.
5. A tiny crystal that bigger ones grow on.

Name: _____